Project Title: Arterial Wall Mechanics

BME 3114: Biomedical Modeling and Simulation Laboratory

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ABSTRACT

Arterial wall mechanics plays a pivotal role in understanding cardiovascular health and disease. This project aims to investigate the mechanical behavior of arterial walls using computational modeling techniques. The arterial wall is a complex structure composed of multiple layers with distinct mechanical properties, including the intima, media, and adventitia. These layers exhibit anisotropic behavior due to the presence of collagen, elastin, and smooth muscle fibers arranged in specific orientations. Fiber orientations derived from a statistical analysis of histological sections from each arterial layer are incorporated into the model. A specific form of the law, requiring only three material parameters for each layer, is employed to investigate the artery's response to combined axial extension, inflation, and torsion. Importantly, the presence of residual stress in an artery in vitro is addressed by assuming that the natural configuration of the material corresponds to an open sector of a tube, which is subsequently closed by initial bending to establish a load-free but stressed circular cylindrical configuration before loading. The report examines the effect of stress within the deformed arterial wall under physiological conditions. The model is calibrated against available arterial data, and its predictions are evaluated for the considered combined loadings.

List of Abbreviations & Symbols

HGO	Holzapfel–Gasser–Ogden

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Chapter 1

Introduction

1.1 Arterial Wall Mechanics

The constitutive framework for arterial wall mechanics, coupled with a comparative study of material models using COMSOL Multiphysics, represents a significant advancement in cardiovascular biomechanics research. Let's discuss into this topic more elaborately. Arteries play a fundamental role in the circulatory system, ensuring proper blood flow to various organs and tissues. The mechanical behavior of arterial walls is critical for maintaining vascular function and responding to hemodynamic forces. Understanding these mechanics is essential for diagnosing and treating cardiovascular diseases like hypertension, atherosclerosis, and aneurysms. The arterial wall is a complex biological structure composed of multiple layers of different tissue types, including elastin, collagen, smooth muscle cells, and endothelial cells. Moreover, arterial tissue exhibits nonlinear, anisotropic, and time-dependent behavior, making its mechanical characterization challenging. Traditional constitutive models may not capture the full complexity of arterial mechanics, necessitating the development of advanced frameworks. The proposed constitutive framework represents a novel approach to modeling arterial mechanics. It aims to address the limitations of existing models by incorporating a more comprehensive representation of arterial tissue behavior. This framework likely integrates advanced constitutive equations derived from experimental data and incorporates multi-scale modeling techniques to capture the hierarchical structure of arterial tissue. The new framework may incorporate elements such as strain energy functions, fiber-reinforced models, or microstructural models to capture the nonlinear and anisotropic behavior of arterial tissue accurately. It may also account for the influence of factors like aging, disease, and mechanical loading on arterial mechanics. Additionally, the framework may offer flexibility in modeling different types of arteries (e.g., elastic vs. muscular) and pathological conditions.

1.2 Histology of Arterial Walls

The histological analysis of arterial walls reveals a complex structure vital for their mechanical function and physiological importance in the circulatory system. Arteries are composed of three distinct layers: the intima, media, and adventitia, each fulfilling crucial roles. The intima, positioned as the innermost layer, consists mainly of endothelial cells supported by a basement membrane and sparse connective tissue. It serves as a crucial barrier between blood flow and the vessel wall, regulating vascular tone and permeability. Beneath the intima lies the media, predominantly comprising smooth muscle cells embedded in a matrix rich in elastin and collagen fibers. This layer imparts elasticity and contractility to arteries, enabling them to adapt to fluctuations in blood pressure and flow. Moving outward, the adventitia, the outermost layer, is characterized by fibroblasts, collagen fibers, and nerve fibers, providing essential structural support and anchoring arteries to surrounding tissues. Arteries contain vasa vasorum, small blood vessels within their walls, facilitating the nourishment of the outer layers. A thorough understanding of the histological organization of arterial walls is crucial for unraveling their mechanical properties and understanding the pathophysiological mechanisms underlying vascular diseases[2].

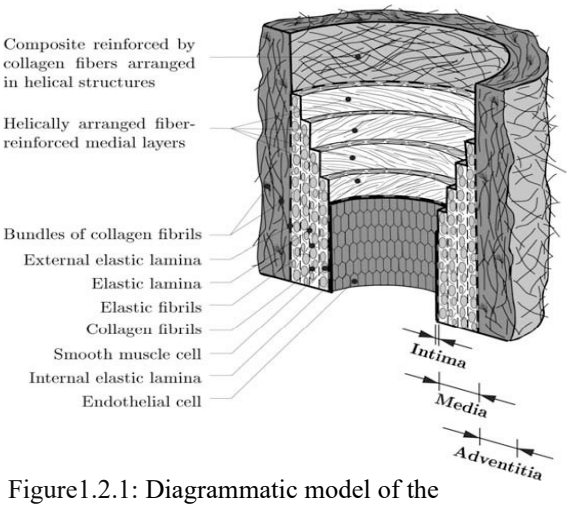


Figure1.2.1: Diagrammatic model of the major components of a healthy elastic artery composed of three layers: intima, media, adventitia[3].

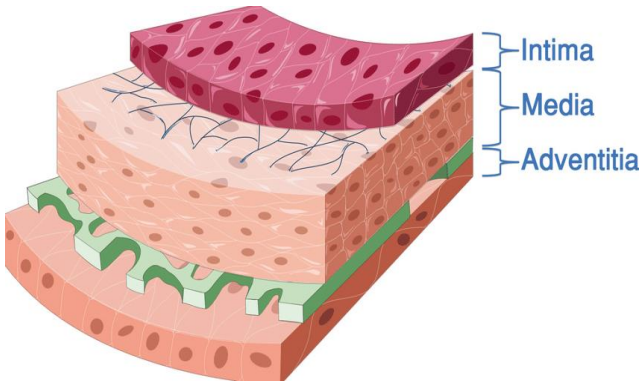


Figure1.2.2: Composition of the arterial wall. The artery wall is composed of three main layers: the intima (internal),), the media (middle), and the adventitia (external)[4].

Intima: The intima, a vital element of arterial walls, constitutes the innermost layer characterized by endothelial cells supported by a basement membrane and sparse connective tissue. It serves as

a critical interface between blood flow and the vessel wall, regulating vascular tone and permeability. The endothelial lining plays a crucial role in maintaining vascular balance by facilitating interactions between circulating blood components and underlying tissues. Furthermore, the intima acts as a protective barrier, thwarting the infiltration of harmful substances and cells into the arterial wall. Dysfunction of the intimal layer, often linked to endothelial dysfunction, inflammation, or injury, is associated with various cardiovascular disorders, such as atherosclerosis and thrombosis. An in-depth comprehension of the complexities of the intimal layer is essential for understanding arterial function and developing strategies for the prevention and treatment of vascular diseases[5].

Media: The middle layer of arteries, known as the tunica media or arterial media, is critical for their strength and function. Made mostly of smooth muscle cells surrounded by connective tissue, this layer controls blood flow and pressure by adjusting the diameter of the artery. It can widen or narrow based on signals from hormones and nerves, adapting blood flow to the body's needs. The arterial media also helps maintain blood pressure by absorbing changes in blood volume and heart output, ensuring a steady supply of blood to organs and tissues[6].

Adventitia: The adventitia is primarily composed of fibroblasts in a loose connective tissue matrix. Growing evidence indicates that, rather than being just a structural support to pulmonary vessels, the adventitia may also play an active role in regulating pulmonary vascular function. The normal adventitia represents approximately 15% of the external diameter of pulmonary arteries larger than 50 μm in diameter, but in IPAH arteries, the adventitial thickness can increase to 25–30% of arterial diameter, predominantly due to collagen deposition. However, on average, the adventitial thickness is not significantly increased in patients with PAH compared to control patients. It is unclear if the increased thickness is due to adventitial extracellular matrix deposition or the accumulation of stromal cells. In PAH, cellular infiltrates of leukocytes, including macrophages, B cells, T cells, and dendritic cells, are seen in the adventitia and likely play a role in the vasoproliferative and vasoconstrictive response[7].

1.3 Mechanical Behavior of Arterial Walls

Methods have been assembled permitting the simultaneous recording of intra-arterial pressure and arterial diameter with sufficient accuracy so that the mechanical properties of arteries in vivo, under physiologic conditions, can be derived. The analysis necessitates finding the relationships between 5 simultaneous variables; hence the data were recorded on magnetic tape and processed by analog and digital computer technics. Analysis of the data established the fact that the mechanical properties of arteries can be described by a linear, first-order differential equation whose coefficients can be defined as the moduli of elasticity and viscosity. Furthermore, the magnitudes of the coefficients were evaluated. The strain which the arteries undergo as a result of arterial pulse pressure variations is normally between 0.01 and 0.04, i.e., between 1 and 4 per cent change in circumference. The total strain associated with marked constriction and dilation does not usually exceed 10 per cent. Therefore, it is established that the circumferential motion of arteries may be characterized as small strain. The fact that the strain is effectively small is confirmed by analytical consideration, i.e., stress-strain relationships can be linearized under physiologic conditions without introducing significant error. The mass of the artery wall does not play a significant role in determining the mechanical behavior of the arteries and can therefore be neglected in any such considerations. The harmonic content, i.e., the frequency spectrum, comprising the pressure and diameter pulsations associated with the cardiac cycle have been evaluated by Fourier analysis.

The magnitudes of the coefficients relating pressure and strain are: elastic (E_p) varies from 1,000 to 6,000 Gm./cm.^{2} (with the exception of a value of 500 in the thoracic aorta of a 12-week-old puppy), viscous (R_p) from 10 to 150, and mass (M_p) from 0.0002 to 0.005. The magnitudes of the coefficients vary from individual to individual, from one site to another within the vascular tree and temporally in response to vasoactive factors. The effects of epinephrine, norepinephrine, acetylcholine and autonomic nerve stimulation on the mechanical properties and behavior of arteries are discussed. The differences between the pressure-strain relationships and the wall tension-strain relationships are discussed, since the mechanical properties of the wall material itself is not given directly by the pressure-strain relationships, i.e., radius and wall thickness must be considered.

An analytical discussion is developed and mathematical equations are derived which relate pressure, wall tension, radius, strain, elasticity, viscosity and inertia in a tube which has been shown

to be a reliable model of the living artery. It is concluded that the concept that the arteries function as a "peripheral heart," i.e., rhythmically contract in synchrony with the heart, is not justified[8].

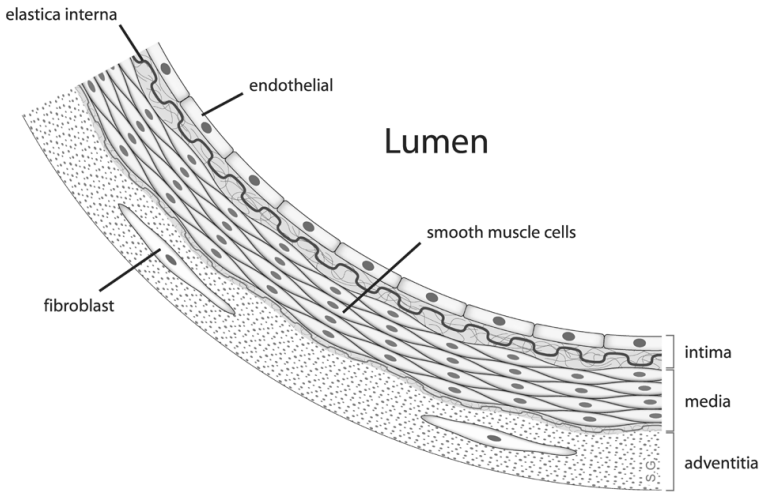


Figure 1.3.1: The anatomy of an artery[9]

1.4 Properties of Arterial Wall Modeling

Hyper Elastic Material:

Hyper elastic materials possess extraordinary elasticity, capable of undergoing significant deformation and returning to their original shape. Exhibiting nonlinear stress-strain behavior, they find applications in diverse fields such as biomedical devices, soft robotics, and sports equipment. Their unique properties offer versatility and resilience in engineering and design solutions.

Anisotropic Nonlinear Properties:

Anisotropic nonlinear properties refer to materials exhibiting varying responses to light or other stimuli depending on directionality. Unlike isotropic materials, which respond uniformly in all directions, anisotropic materials exhibit different optical behaviors along different axes. These properties arise from asymmetrical molecular structures or crystal lattices, leading to directional variations in refractive index, birefringence, and nonlinear optical effects. Anisotropic nonlinear materials find applications in advanced optical devices such as modulators, switches, and

frequency converters. Understanding and engineering these properties are crucial for developing cutting-edge technologies in photonics, telecommunications, and quantum computing.

Artery Hyper elasticity:

Artery hyperelasticity refers to the ability of arteries to stretch significantly under pressure and return to their original shape. This property is essential for maintaining blood flow dynamics and protecting against damage from pulsatile blood flow. Arterial walls, composed of elastic fibers and smooth muscle cells, undergo reversible deformation due to changes in blood pressure. Hyperelastic behavior ensures efficient nutrient delivery and waste removal to tissues. However, pathological alterations in arterial elasticity can lead to hypertension, atherosclerosis, and aneurysms. Understanding arterial hyperelasticity aids in developing therapies to mitigate cardiovascular diseases and designing biomimetic materials for medical implants and vascular tissue engineering[10].

Viscoelasticity:

Viscoelasticity in arteries refers to their ability to deform under pressure and return to their original shape. This property enables efficient blood flow regulation, absorbing pulsatile energy and maintaining vascular integrity. Understanding arterial viscoelasticity is crucial in diagnosing and treating cardiovascular diseases like hypertension and atherosclerosis.

Five-branch Viscoelasticity:

Five-branch viscoelasticity is a model used to represent the time-dependent mechanical behavior of materials, particularly those that exhibit both elastic (immediate response) and viscous (delayed or time-dependent) characteristics. In this context, it's applied to simulate the dynamic response of an artery wall, specifically the media layer. Here's a breakdown of the key aspects:

Five branches viscoelasticity model:

Five branches: The model comprises five independent "branches," each acting like a mini-model within the overall system.

Spring and Dashpot: Each branch likely consists of two elements connected in parallel:

Spring: Represents the elastic component of the material. When a force is applied, the spring deforms immediately, storing energy that's released when the force is removed.

Dashpot: Represents the viscous component of the material. It acts like a damper, resisting deformation but dissipating energy as heat over time.

Relaxation Times: Each branch has a unique relaxation time. This refers to the time it takes for the stress within the dashpot to decay to a specific fraction (usually 1/e or about 37%) of its initial value after the stress is applied. The five branches have different relaxation times, allowing the model to capture a range of time-dependent behaviors.

1.5 HGO Model

The Holzapfel-Gasser-Ogden (HGO) model is one of the most popular anisotropic hyper elastic models. Anisotropic hyper elasticity is also a useful building block for non-linear visco plastic material models. The original HGO model was developed for anisotropic arteries, where the material is represented using 2 families of collagen fibers embedded into an isotropic matrix [10].

$$\Psi(F) = \Psi_{iso}(F) + \Psi_{aniso}(F, a_1, a_2) \tag{1}$$

The Holzapfel-Gasser-Ogden (HGO) model for anisotropic hyper elastic behavior of collagen fiber reinforced materials was initially developed to describe the elastic properties of arterial tissue, but is now used extensively for modelling a variety of soft biological tissues. Such materials can be regarded as incompressible, and when the incompressibility condition is adopted the strain energy Ψ of the HGO model is a function of one isotropic and two anisotropic deformation invariants. A compressible form (HGO-C model) is widely used in finite element simulations whereby the isotropic part of Ψ is decoupled into volumetric and isochoric parts and the anisotropic part of Ψ is expressed in terms of isochoric invariants. Here, by using three simple deformations (pure dilatation, pure shear and uniaxial stretch), we demonstrate that the compressible HGO-C formulation does not correctly model compressible anisotropic material behavior, because the anisotropic component of the model is insensitive to volumetric deformation due to the use of isochoric anisotropic invariants. In order to correctly model compressible anisotropic behavior we present a modified anisotropic (MA) model, whereby the full anisotropic invariants are used, so that a volumetric anisotropic contribution is represented. The MA model correctly predicts an anisotropic response to hydrostatic tensile loading, whereby a sphere deforms into an ellipsoid. It

also computes the correct anisotropic stress state for pure shear and uniaxial deformations. To look at more practical applications, we developed a finite element user-defined material subroutine for the simulation of stent deployment in a slightly compressible artery. Significantly higher stress triaxiality and arterial compliance are computed when the full anisotropic invariants are used (MA model) instead of the isochoric form (HGO-C model) [11].

Chapter 2

Motivation/ Background story

In the last few years there has been a significant growth in interest in the mechanical properties of biological soft tissue treated from the continuum mechanical perspective, and, in particular, in the mechanics of arterial tissue. The project on "Modeling and Simulation of Arterial Wall Mechanics by using COMSOL Multiphysics" likely originated from a pressing need to understand the intricate dynamics of arterial walls and their impact on cardiovascular health. Arterial diseases like atherosclerosis, hypertension, and aneurysms are major causes of morbidity and mortality worldwide. However, the underlying mechanics governing these conditions are complex and not fully understood. Motivated by this challenge, the project aims to develop a comprehensive computational model using COMSOL Multiphysics, a powerful simulation software. This model could provide insights into phenomena such as wall deformation, stress distribution, and the formation of plaques or lesions. Understanding arterial wall mechanics at a nuanced level could revolutionize the diagnosis, treatment, and prevention of cardiovascular diseases. Clinicians could use the simulations to predict patient-specific responses to interventions like stent placement or medication, leading to more personalized healthcare strategies. Moreover, insights gained from the project could inspire advancements in medical device design and pharmacological approaches, ultimately enhancing patient outcomes and reducing the global burden of cardiovascular disease.

Chapter 3

Methodology

This methodology outlines the process of simulating arterial wall mechanics using the Holzapfel-Gasser-Ogden (HGO) constitutive framework, as proposed by Holzapfel, Gasser, and Ogden (2000), and implemented in the COMSOL Multiphysics model "Arterial Wall Mechanics." The goal is to accurately capture the anisotropic and hyper elastic behavior of arterial tissues under physiological loading conditions**.**

It also outlines dynamic behavior of the artery is studied, especially the viscoelastic response is calculated by adding a five branches viscoelasticity feature to the hyper elastic material.

3.1 Mathematical equations of HGO Model

The HGO model is an incompressible anisotropic hyperelastic material model defined by an isochoric strain energy density.

The incompressibility condition implies adding a volumetric stress, S_{vol}

$$S_{vol} = -p_w J c^{-1} \quad (1)$$

Here, J represents the determinant of the deformation gradient tensor, F.

The auxiliary pressure p_w ensures the incompressibility with the weak equation:

$$\text{Weak}=(J-1)\ \text{test}(p_w) \quad (2)$$

The isochoric strain energy density is defined by a function of the form:

$$W_s = W_1 + W_4 + W_6 \quad (3)$$

The three terms on the right-hand side of Equation 3 depend on invariants of the **right Cauchy-Green tensor.**

The first term describes the mechanical behavior of the elastic ground substance. The isotropic function W1 depends on one material parameter and the first isochoric invariant $I_1\overline{C_{el}}$, defined in the same fashion as for a neo-Hookean material.

$$W_1 = \frac{c}{2}(I_1(\overline{C_{el}}) - 3) \quad (4)$$

Here, $I_1\overline{C_{el}}$, represents the average value of the first invariant of the right Cauchy-Green deformation tensor C over an element or a material region. In a finite element analysis or continuum mechanics context, $\overline{C_{el}}$ refers to the right Cauchy-Green deformation tensor calculated at the element level. This tensor characterizes the deformation within each finite element of the mesh.

The second and third terms on the right-hand side of Equation 3 describe the mechanical contribution of the collagen fiber network. These expressions are written as:

$$W_4 = \frac{k_1}{2k_2}(e^{k_2}(I_4 - 1)^2 - 1) \quad (5)$$

$$W_6 = \frac{k_1}{2k_2}(e^{k_2}(I_6 - 1)^2 - 1) \quad (6)$$

This comes from the "**Strain-Energy Function**" Proposed by Delfino et al []. They proposed an (isotropic) rubber-like potential for carotid arteries which is able to model the typical stiffening effects in the high pressure domain. The strain-energy function $\bar{\psi}$ has the form:

$$\bar{\psi} = \frac{a}{b}\{\exp\left[\frac{b}{2}(\bar{I}_1 - 3)\right] - 1\} \quad (7)$$

where a > 0 is a stress-like material parameter and b > 0 is a non-dimensional parameter. The first invariant of the modified right Cauchy–Green tensor $\bar{C}$ is defined as $\bar{I}_1 = \bar{C}$: I. Since the exponential function increases monotonically with $\bar{I}_1$ it is easy to show that strict local convexity of the potential (7) as a function of $\bar{C}$.

In (6), the fiber network is reduced to two families of fibers with material properties k_1 and k_2.

Studying the dynamic behavior of the artery and calculate the time-dependent response given a sudden axial stretching.

The two modeled layers (media and adventitia) are described by the anisotropic HGO hyperelastic material model:

$$W_s = W_1 + W_4 + W_6 = W_{iso} \quad (8)$$

Only the media layer includes viscoelastic behavior. The generalized Maxwell viscoelastic model is used to represent relaxations time at different time-scales.

$$S = S_{iso} + S_{vol} \sum_m q_m \quad (9)$$

This equation is commonly used in continuum mechanics.

The equation suggests that the total stress (*S*) is the sum of the isochoric stress (*S*iso) and the volumetric stress (*S*vol), along with the contributions from individual stress components (qm).

For each branch of the generalized Maxwell model, the viscoelastic stress follow:

$$Q_m + \frac{Q_m}{\tau_m}^{\cdot} = \beta_m S_{iso}^{\cdot} \quad (10)$$

where τ_m is the relaxation time and β_m is the energy factor per branch. Applying a variable change $q_m = \beta_m S_{iso} - Q_m$ gives:

$$\tau_m \, q_m^{\cdot} + q_m = \beta_m S_{iso} \quad (11)$$

In this arterial wall model, a generalized Maxwell viscoelastic model with five branches is used.

The artery is first loaded with an internal pressure of 100 mmHg and an initial axial stretch of 1.5. After initialization, the stretch is increased to 1.7 and the viscoelastic relaxation is calculated.

3.2 Design

3.2.1 Model Parameters

The artery wall mechanics model in this study is defined by several key parameters that describe the geometry, material properties, and boundary conditions of the arterial wall. These parameters are essential for accurately representing the mechanical behavior of the artery under different physiological conditions. In this section, e a detailed description of the model parameters and their significance in the artery wall mechanics model is provided.

The HGO model is an incompressible anisotropic hyper-elastic material model defined by an isochoric strain energy density. Following Ref. [21], the media and adventitia are modeled as a layered cylindrical tube. Model symmetry allows the use of 2D axisymmetric model. Typically mechanical experiments measure the response of arterial sections subject to combined axial stretch and internal blood pressure.

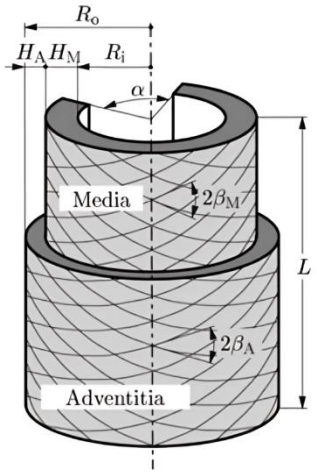


Figure 3.1: Graphical representation of different parameters of the artery model

Table 3.1: List of parameters mentioning their values, units and description

Parameter	**Value**	**Unit**	**Description**
HM	0.26	mm	Thickness of the Media layer
HA	0.13	mm	Thickness of the Adventitia layer
Ri	0.71	mm	Inner Radius of the Artery
Ro	Ri+HM+HA	mm	Outer Radius of the Artery
sector_angle	10	degree	Sector Angle refers to the Relative angle between the two fiber families (a01 and a02) within the arterial wall.
L	2.5	mm	Initial artery length
rhoM	1100	kg/m^3	Density of the Media layer
rhoA	1100	kg/m^3	Density of the Adventitia layer
cM	3	kPa	Lamé parameter of Media layer (HGO material parameter)
cA	0.3	kPa	Lamé parameter of Adventitia layer (HGO material parameter)
k1M	2.3632	kPa	Fiber stiffness of Media layer (HGO material parameter)
k1A	0.5620	kPa	Fiber stiffness of Adventitia layer (HGO material parameter)
k2M	0.8393	-	Model parameter of Media layer (HGO material parameter)
k2A	0.7112	-	Model parameter of Adventitia layer (HGO material parameter)
k3M	0	-	Fiber dispersion of Media layer (HGO material parameter)
k3A	0	-	Fiber dispersion of Adventitia layer (HGO material parameter)
betaM	29.0	degree	Angle between fibers in the Media layer
betaA	62.0	degree	Angle between fibers in the Adventitia layer
p_i	160	mmHg	Internal pressure
lambda_z	1.5	-	Initial Axial stretch

All the data used for implementing this project was collected from ref. [16].

3.2.2 Geometry

In COMSOL Multiphysics, the geometry setup is a critical step for modeling accurately. A very simple geometry is used for implementing artery wall model which involves a rectangle with a width of (Ro-Ri) and height of L. A layer was added to the right which has the thickness of HA (thickness of outer adventitia layer).

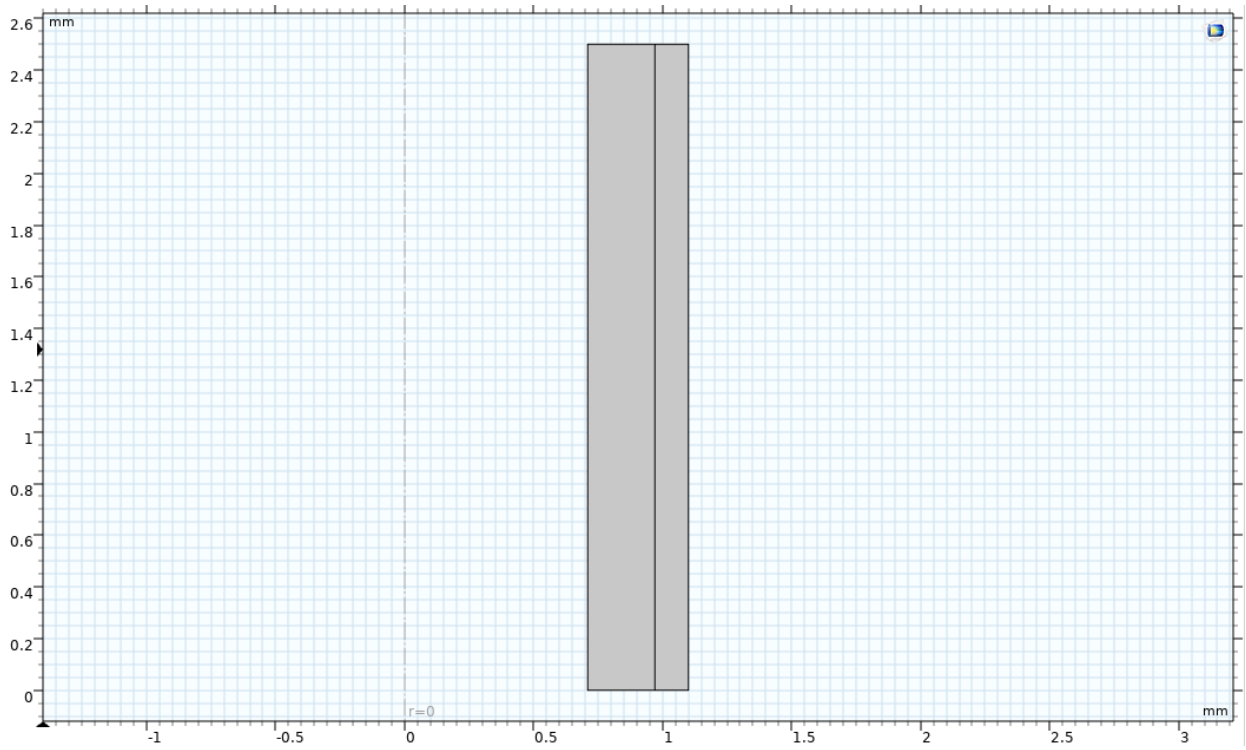


Figure 3.2: Simple 2D Geometry of the artery model

3.2.3 Materials

The HGO model captures the anisotropic, nonlinear mechanical response observed in experiments on excised arteries. The key features of this material model are:

- **Hyperelasticity:** The HGO model describes arterial behavior within the theory of hyperelasticity. It accounts for large elastic deformations.
- **Anisotropy:** The model considers the anisotropic properties of collagen fibers in the media and adventitia.

- **Incompressibility:** Since arteries do not change their volume within the physiological range of deformation [4], they can be regarded as incompressible materials, so the HGO model ensures incompressibility by adding a volumetric stress term.

From ref [16] [21], The material parameters for the media and the adventitia are given in the following table

Table 3.2: Material Parameters used for the media and the adventitia layer

Material Property	**Value (Media)**	**Value (Adventitia)**	**Units**
Density (rho)	1100	1100	kg/m^3
Lamé parameter (c)	3	0.3	kPa
Fiber stiffness (k1)	2.3632	0.5620	kPa
Model parameter (k2)	0.8393	0.7112	-
Fiber Dispersion (k3)	0	0	-
Angle between fibers within the layer (β)	29.0	62.0	degree

- **Lamé parameter (c)** indicates the shear deformation. Shearing forces cause shearing deformation. An element subject to shear does not change in length but undergoes a change in shape.
- **Fiber stiffness (k1)** refers to the hardness of the layers of the artery.
- **Model parameter (k2)** refers to HGO Model Parameter
- **Fiber dispersion (k3)** refers to the arrangement and distribution of collagen fibers within the arterial wall.

3.2.4 Solid Mechanics

The symmetry boundary condition allows the bottom end of the artery to freely expand in the radial direction. On the top surface, prescribed displacements in the axial direction account for the axial stretching. Internal pressure is applied with a pressure boundary load on the inner surface.

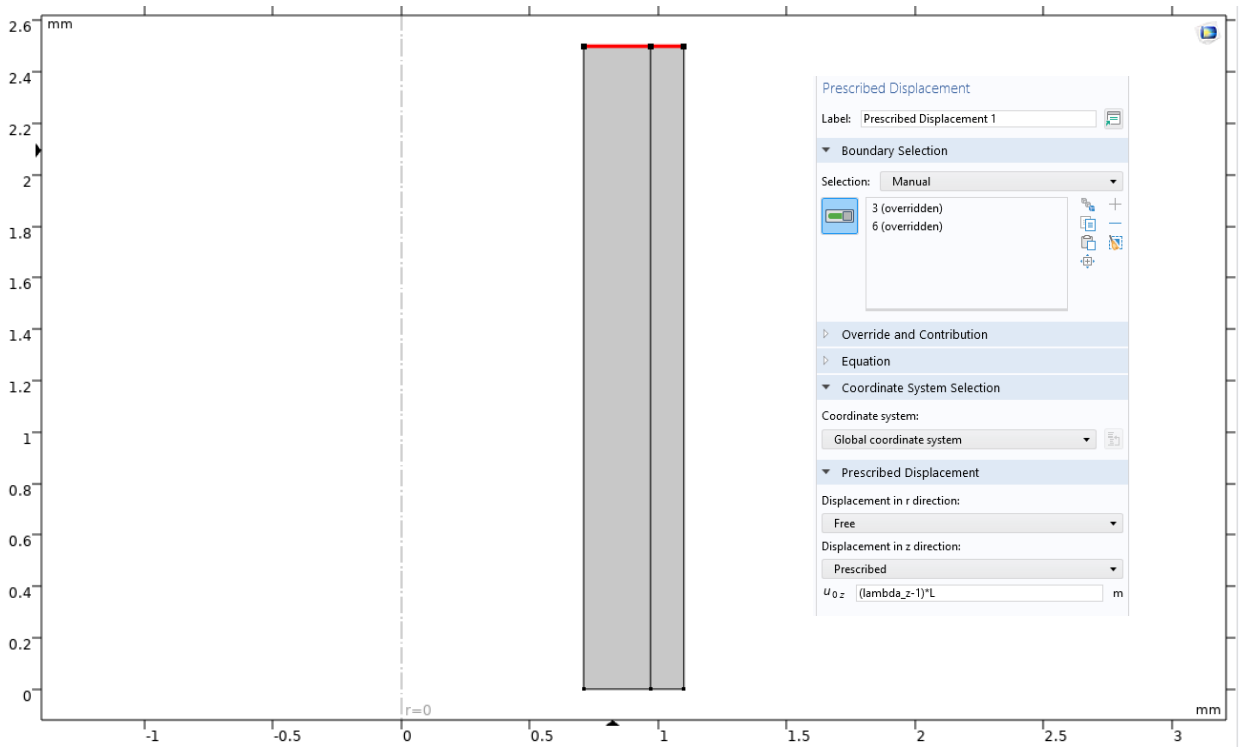


Figure 3.3: Prescribed displacement applied on boundary 3 and 6

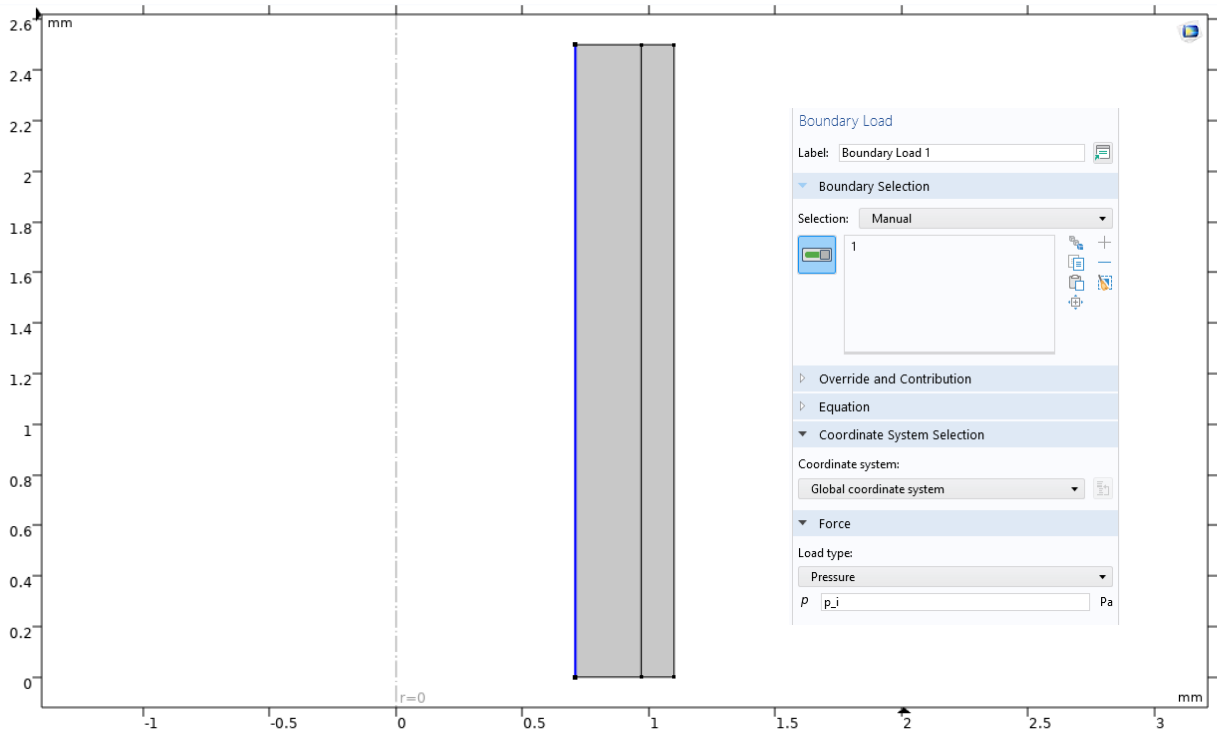


Figure 3.4: Boundary Load applied on boundary 1

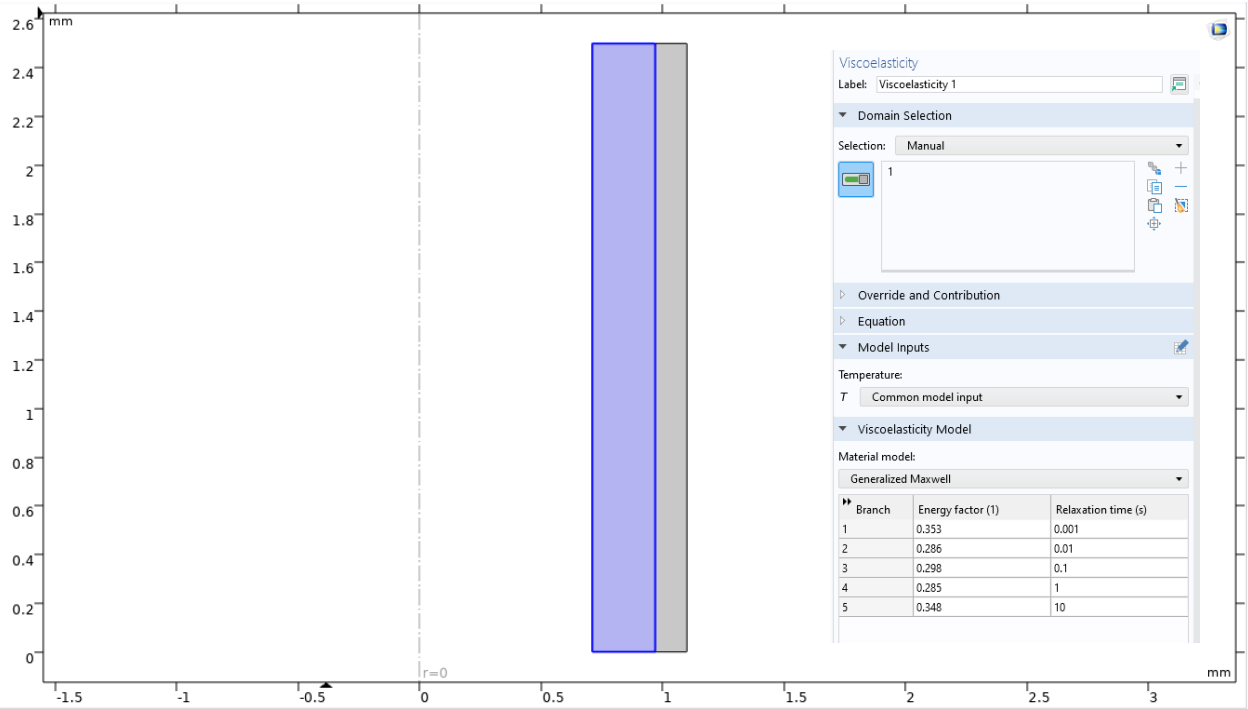


Figure 3.5: For viscoelasticity, selected domain 1 and the energy factor and relaxation time for five branches

3.2.5 Hyperelastic Material

In arterial wall mechanics simulation, adding fiber families is essential because it allows for a more accurate representation of the complex structure and mechanical behavior of the arterial wall.

Hyperelastic Material (Media): Domain 1 was selected for Hyperelastic Material (Media), then two fiber families were inserted as fiber family 1 and fiber family 2. For both of the fiber families.

Table 3.3: Fiber Model of Hyperelastic Material (Media) fiber families

Fiber Family 1	Coordinate System	Rotated System Media Fiber Family 1 (Rotation angle β = betaM (29 degree))
	Material Model	Holzapfel-Gasser-Ogden
	Material	Domain Material
Fiber Family 2	Coordinate System	Rotated System Media Fiber Family 2 (Rotation angle β = - betaM (-29 degree))
	Material Model	Holzapfel-Gasser-Ogden
	Material	Domain Material

Hyperelastic Material (Adventitia): Domain 2 was selected for Hyperelastic Material (Adventitita), then two fiber families were inserted as fiber family 1 and fiber family 2. For both of the fiber families.

Table 3.4: Fiber Model of Hyperelastic Material (Adventitia) fiber families

Fiber Family 1	Coordinate System	Rotated System Adventitia Fiber Family 1 (Rotation angle β = betaA (62 degree))
	Material Model	Holzapfel-Gasser-Ogden
	Material	Domain Material
Fiber Family 2	Coordinate System	Rotated System Adventitia Fiber Family 2 (Rotation angle β = - beta (-62 degree))
	Material Model	Holzapfel-Gasser-Ogden
	Material	Domain Material

3.2.6 Viscoelastic Model

For viscoelastic model, a generalized Maxwell viscoelastic model with five branches is used with the following values taken from ref. [15][17]:

Table 3.5: Table for Energy factors and Relaxation times used for five branches of viscoelastic model

Branch	Energy factor (1)	Relaxation time (s)
1	0.353	0.001
2	0.286	0.01
3	0.298	0.1
4	0.285	1
5	0.348	10

3.2.7 Mesh

Using mapped mesh in artery wall mechanics simulations allows for accurate representation of the complex geometry and material properties of the arterial wall. Mapped meshing involves dividing the geometry into elements that closely follow its shape and contours, which improved the accuracy of the simulation results, especially in regions of interest such as the adventitia and media

layers of the artery. At distribution 1, number of elements = 6 and at distribution 2, number of elements = 4. Since distribution 1 represents Media layer, it has more elements because media layer is typically thicker and more elastic compared to the adventitia.

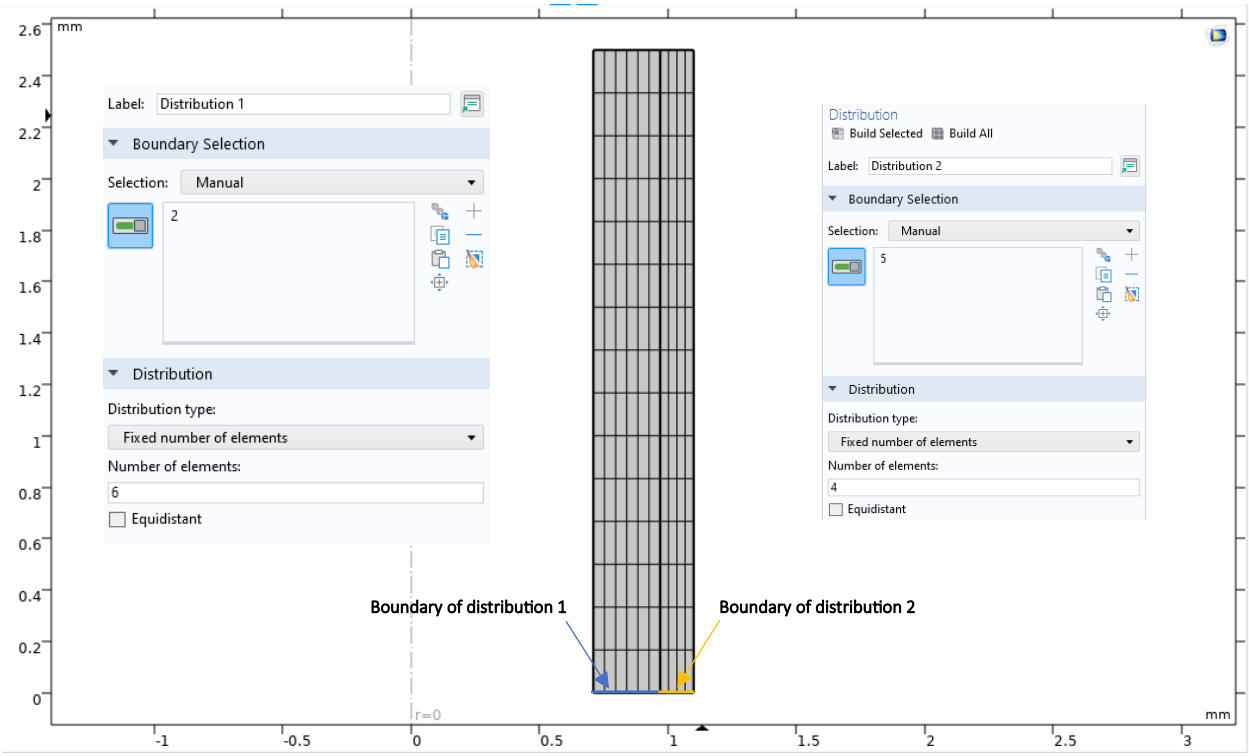


Figure 3.6: Mesh structure of the artery model

3.2.8 Sector Revolution

Sector Revolution was applied for both Adventitia and Media layer was applied.

For Media Sector Revolution,

Start Angle = - 40 degree

Revolution Angle = 140 degree

For Adventitia Sector Revolution,

Start Angle = -50 degree

Revolution Angle = 160 degree

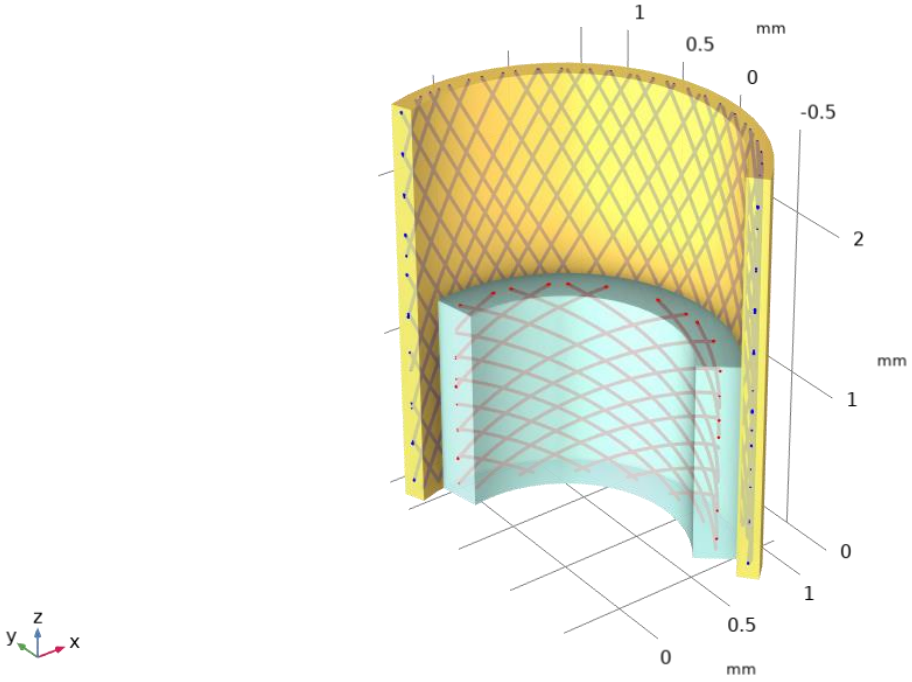


Figure 3.7: Complete model of the Artery wall containing Media and Adventitia layer with fiber families

Chapter 4

Results & Discussion

4.1 Stationary Results:

These results depict the initial state of the artery after it's been pre-stressed with a specific internal pressure and an initial axial stretch. They are obtained from Study 1 (Stationary), which doesn't consider the viscoelastic behavior, rather essentially simulates how the artery responds to the initial load without taking into account the time-dependent stress relaxation.

Here's what we observed in the stationary results:

1. Pressure vs. Radius: From the COMSOL model, we investigated the relationship between axial pre-stretch, internal pressure, and inner radius in a carotid artery model for three different axial pre-stretch values i.e,

- Lamda_z =1.5 for 50% increase in length
- Lamda_z =1.7 for 70% increase in length
- Lamda_z =1.9 for 90% increase in length.

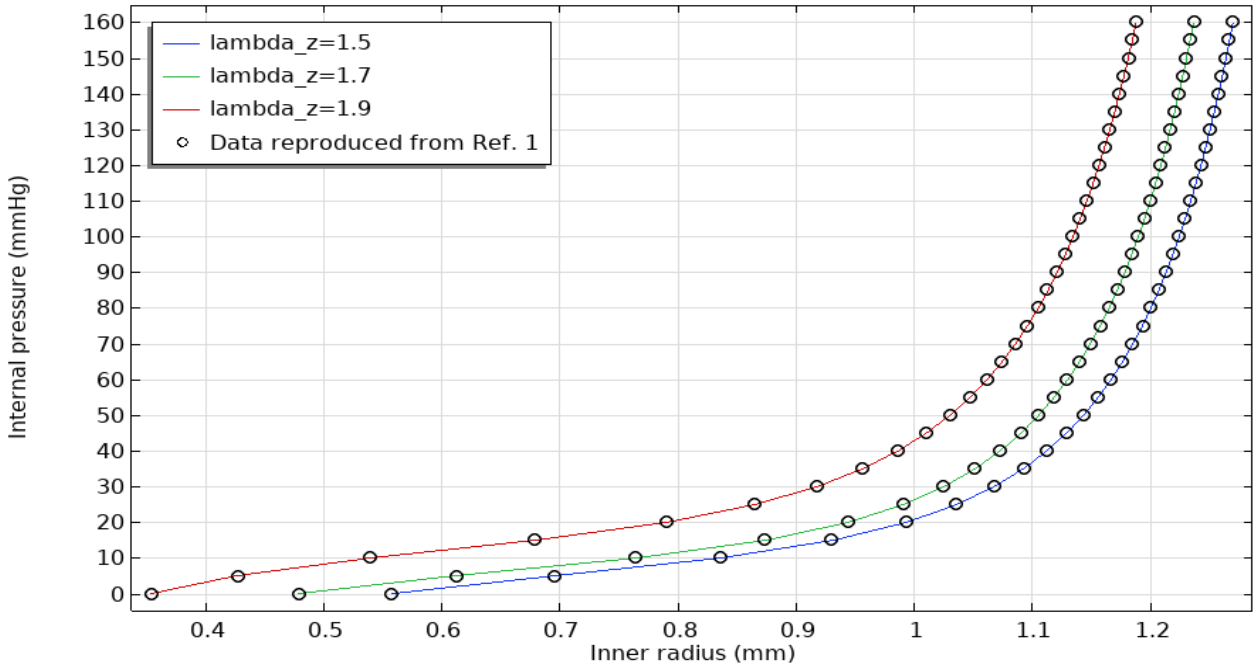


Figure 4.1: Internal pressure (mmHg) vs Inner radius (mm) plot

Initial Linear Region: At low pressures, the pressure-radius relationship is nearly linear. This can be attributed to the initial uptake of load by the elastin fibers within the arterial wall. Elastin fibers are known for their high extensibility and provide a relatively linear response to low stress.

Exponential Stiffening: As the pressure increases, the wavy collagen fibers in the arterial wall begin to straighten and take up more load. This progressive recruitment of collagen fibers leads to a significant increase in stiffness, reflected by the exponential rise in pressure.

Table 4.1 : Comparison of few data points extracted from figure.

Pressure (mmHg)	Radius(mm) (lambda_z=1.5)	Radius(mm) (lambda_z=1.7)	Radius(mm) (lambda_z=1.9)
80	1.19	1.16	1.10
100	1.22	1.18	1.13
120	1.24	1.21	1.15
140	1.26	1.22	1.17

From the table 4.1 **,** we can observe that the inner radius increases as the internal pressure increases for all three axial stretches. However, the rate at which the artery widens depends on the axial stretch.

At lower axial stretches (more relaxed artery): The initial portion of the curve will be steeper because the compliant wall can bulge outwards more readily with increasing pressure.

At higher axial stretches (stiffer artery): The curve will be shallower from the beginning because the stiffer wall offers more resistance to expansion even at lower pressures

Thus it can be observed , at lower axial stretches (lambda_z = 1.5), the inner radius increases more compared to higher stretches (lambda_z = 1.9) for the same pressure increment. A higher axial stretch (lambda_z = 1.9) represents a stiffer artery (possibly due to increased muscle tone or changes in the collagen structure) , similar to pulling a balloon tighter. A stiffer material resists deformation more. The stiffer the artery (higher axial stretch), the less it widens under pressure.

4.2 Radial stress:

Pressure:0mmHg

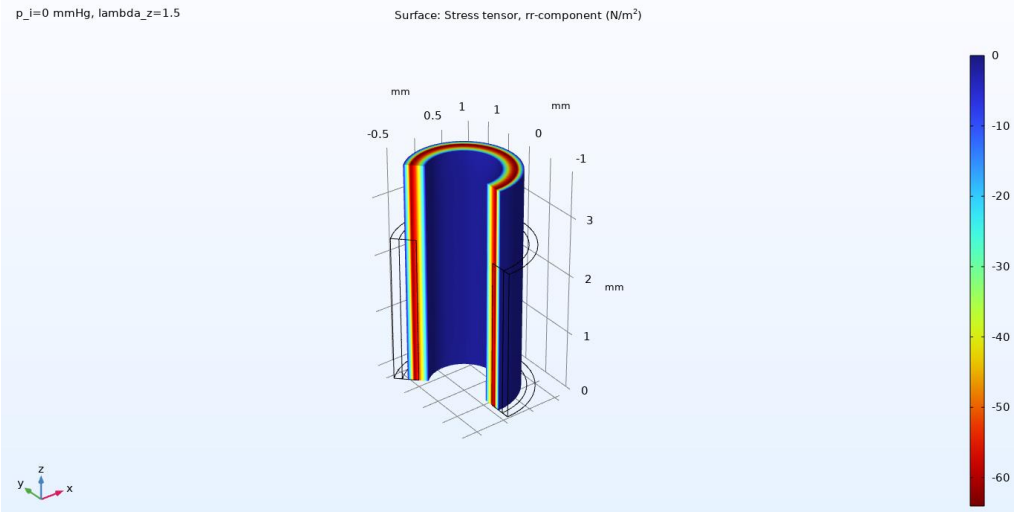


Figure 4.2.1: Simulated plot for 0mmHg of pressure and lambda_z=1.5

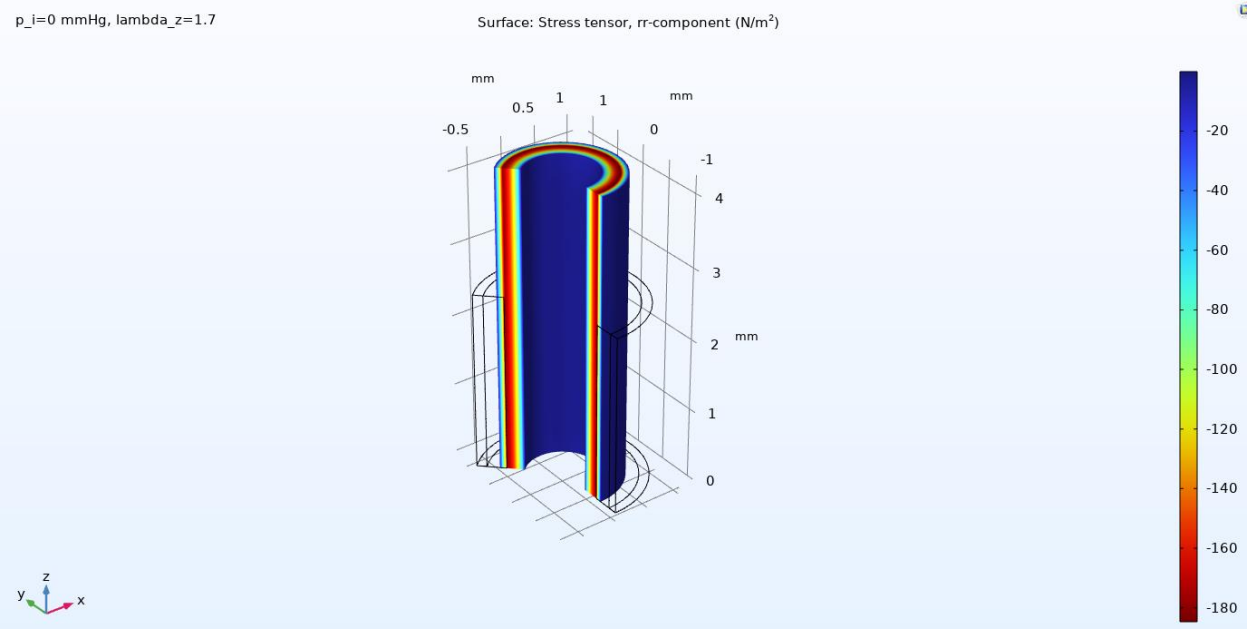


Figure 4.2.2: Simulated plot for 0mmHg of pressure and lambda_z=1.7

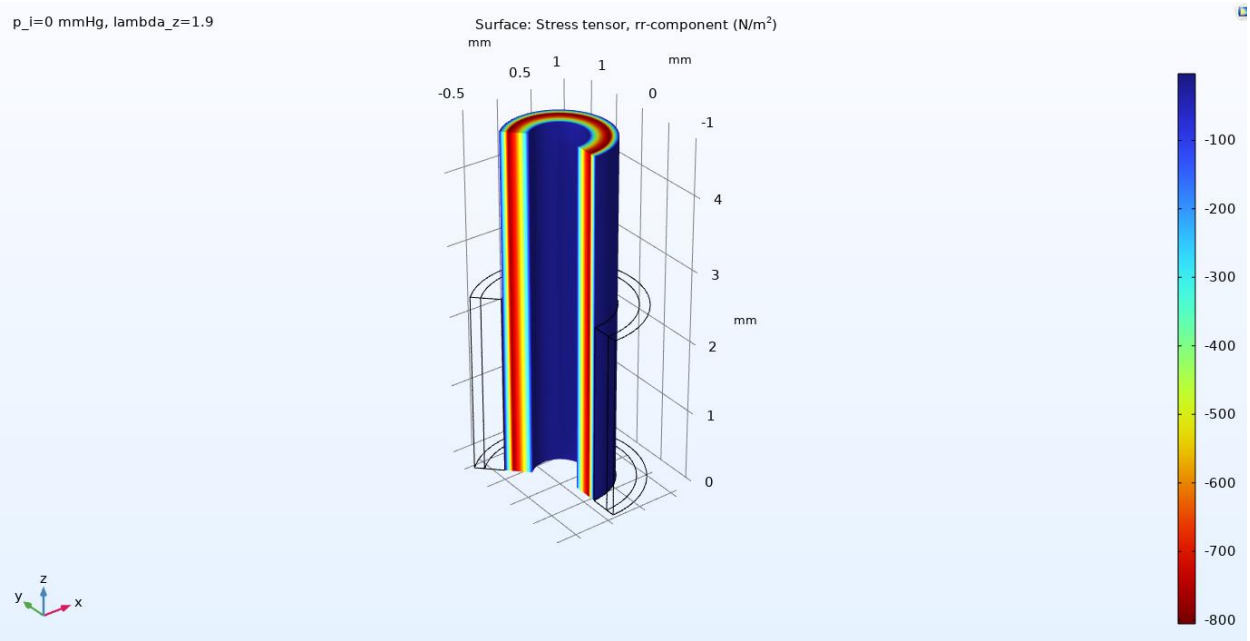


Figure4.2.3: Simulated plot for 0mmHg of pressure and lambda_z=1.9

Pressure:80mmHg

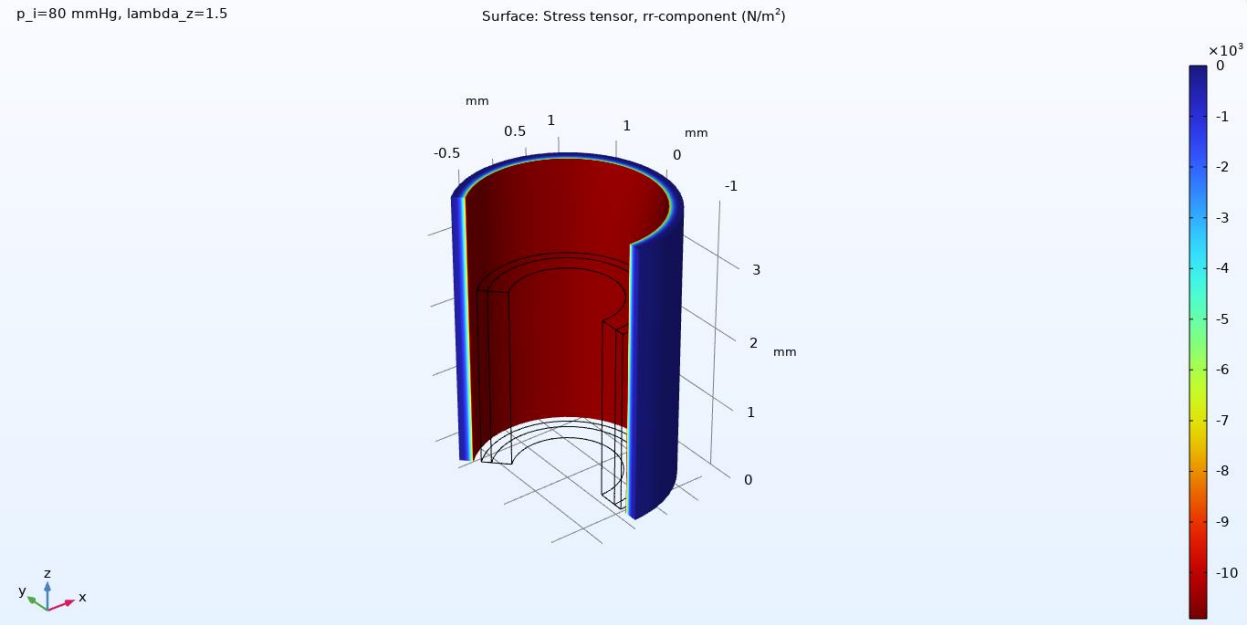


Figure4.2.4: Simulated plot for 80mmHg of pressure and 1.5 of lambda_z

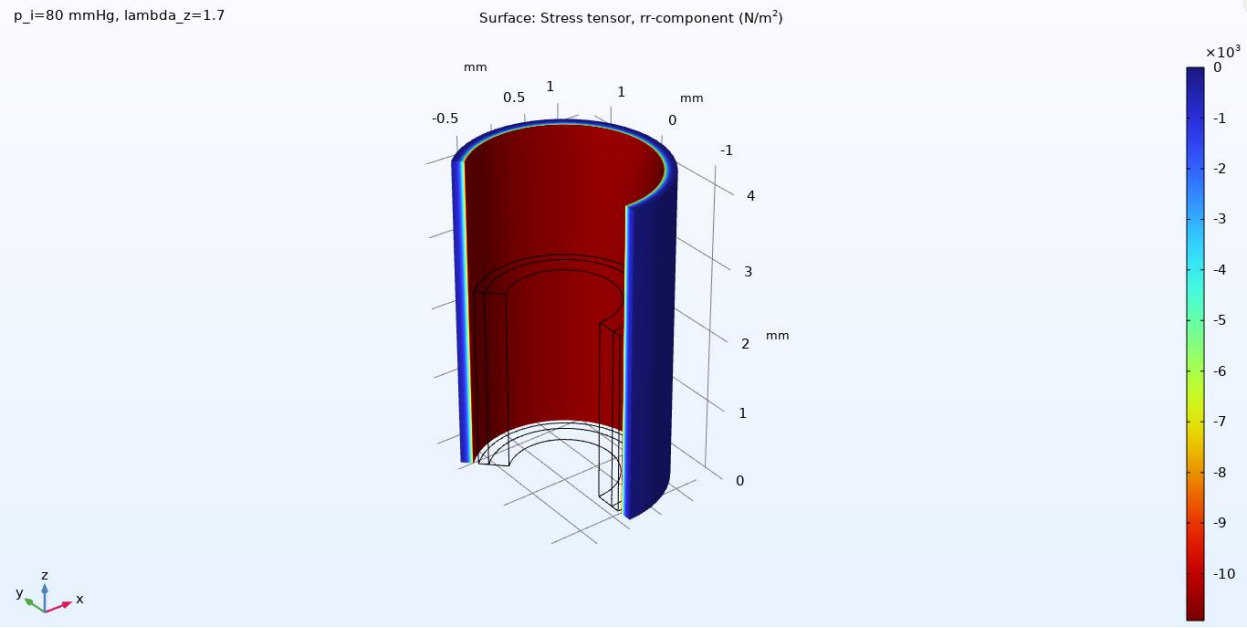


Figure4.2.5: Simulated plot for 80mmHg of pressure and lambda_z=1.7

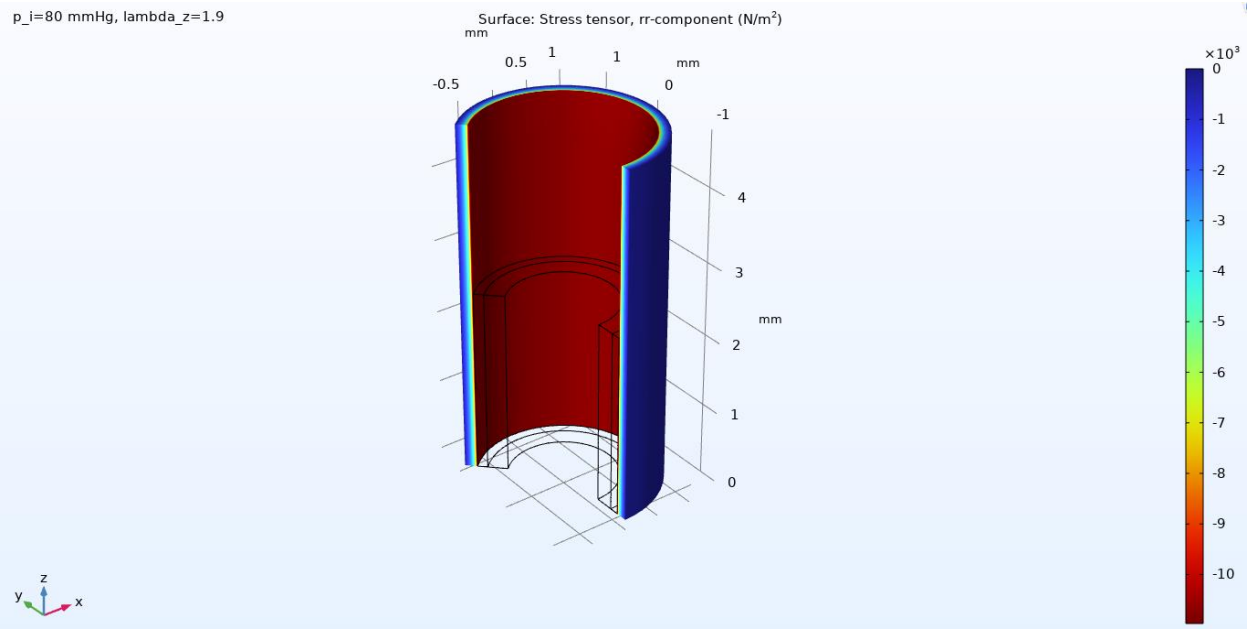


Figure4.2.6: Simulated plot for 80mmHg of pressure and lambda_z=1.9

Pressure:160mmHg

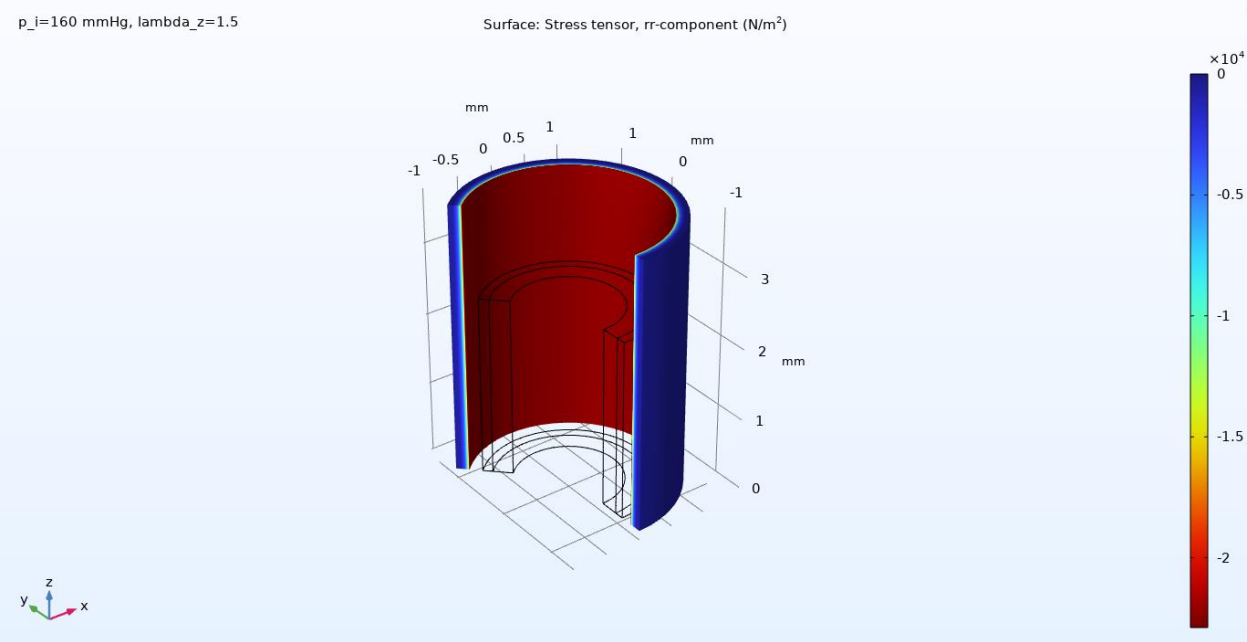


Figure4.2.7: Simulated plot for 160mmHg of pressure and lambda_z=1.5

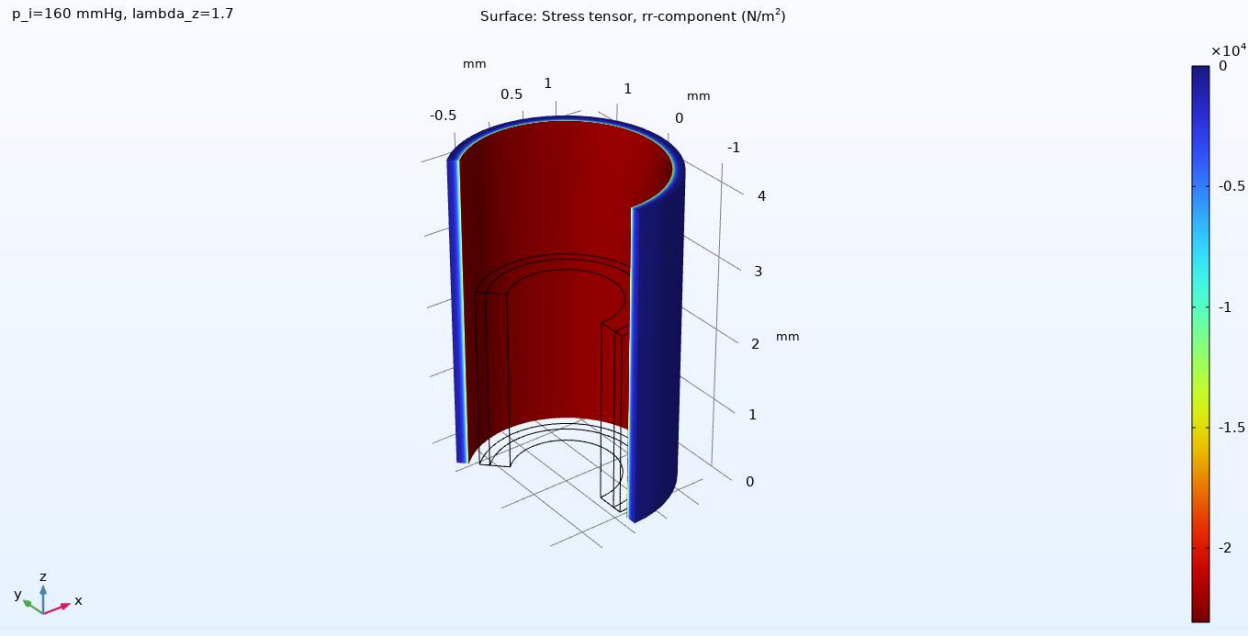


Figure 4.2.9: Simulated plot for 160mmHg of pressure and lambda_z=1.7

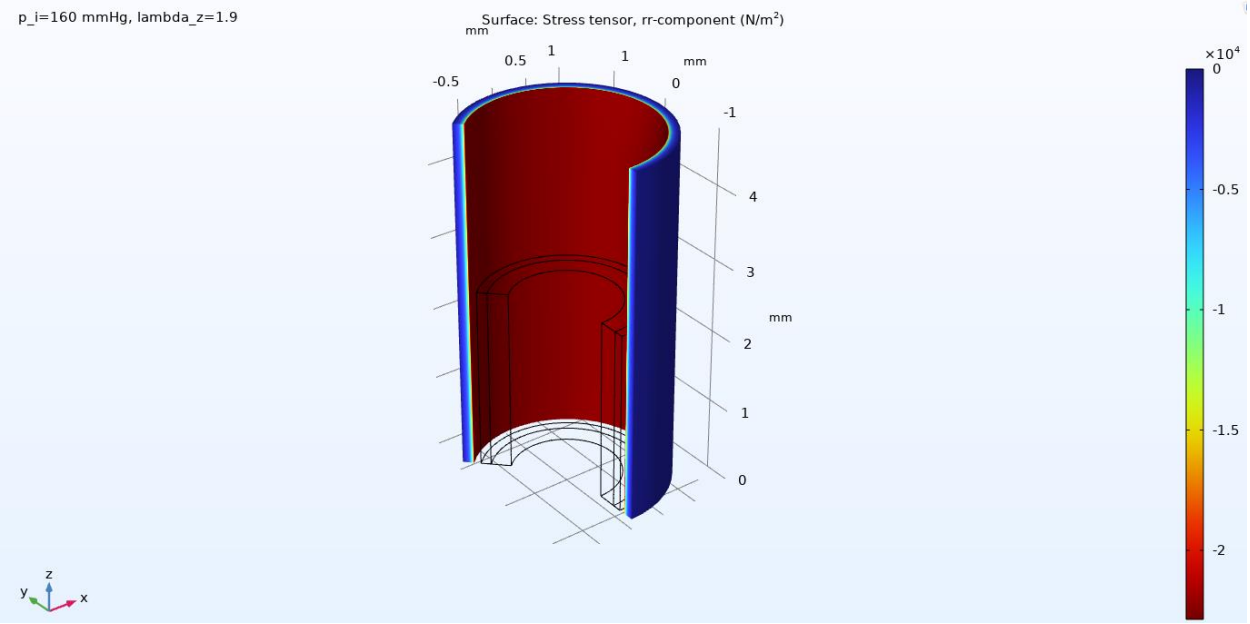


Figure4.2.10: Simulated plot for 160mmHg of pressure and lambda_z=1.9

Factors Affecting Radial Stress:

Axial Stretch (lambda_z): As the axial stretch increases (1.5 to 1.9), the artery will become stiffer. A stiffer artery will experience a different stress distribution compared to a more relaxed artery under the same pressure. You would likely see a smaller overall change in radial stress across the arterial wall thickness for higher axial stretches.

Internal Pressure (P): As the internal pressure increases (0 mmHg to 160 mmHg), the radial stress will generally increase throughout the arterial wall thickness. This is because the pressurized blood pushes outwards, exerting a greater force that needs to be balanced by the radial stress within the wall. The color gradient would likely shift towards higher stress values as pressure increases.

Color Gradient Interpretation:

In COMSOL, the color gradient of a stress plot typically represents the magnitude of the stress. We can observe from the change in color gradient:

- Red: Highest radial stress (tensile or compressive)
- Blue: Lowest radial stress (tensile or compressive)
- Green/Yellow: Intermediate values of radial stress

Effect of Axial Stretch: At a constant pressure (e.g., 0 mmHg):

Lower axial stretch (1.5): Wider range of colors in the gradient, indicating a larger variation in radial stress across the wall thickness.

Higher axial stretch (1.9): Narrower range of colors, indicating a smaller variation in radial stress due to the stiffer wall.

Effect of Internal Pressure: Considering the same axial stretch:

Lower pressure (0 mmHg): The entire color gradient shifts towards the "blue" end, indicating lower overall radial stress in the inner wall.

Higher pressure (160 mmHg): The gradient shifts towards the "red" end, indicating higher overall radial stress throughout the inner wall.

4.3 Viscoelasticity Results:

These results capture the dynamic behavior of the artery when an additional stretch is applied over time. They are obtained from Study 2 (Time Dependent), which incorporates the viscoelastic model of the media layer and simulates how the stress in the artery relaxes over time due to the viscoelastic properties. Here's what we observed in the viscoelasticity results:

1. **Force Relaxation**: This plot shows how the total force exerted by the artery decreases over time after the additional stretch is applied. This decrease reflects the relaxation of viscoelastic stress.

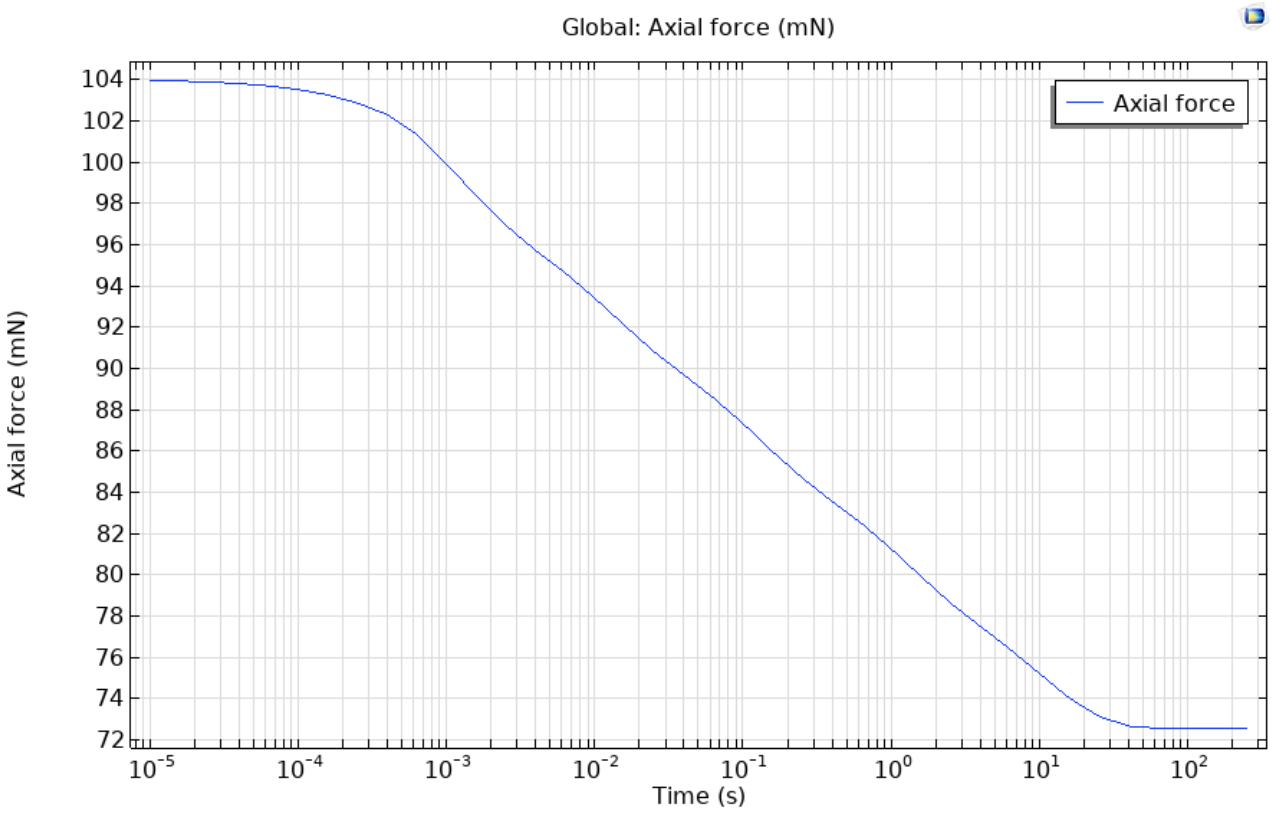


Figure 4.3.1:Axial force (Nm) vs Time (sec) plot

The model simulates the artery's response to a change in axial stretch and axial force with time. The artery is initially loaded with an internal pressure of 100 mmHg and stretched to a length 1.5 times its original length. After this initial state, the stretch is further increased to 1.7, and the model

calculates the resulting viscoelastic relaxation (stress decrease over time). The total force exerted by the artery is computed and plotted in Figure 4.2 .The results show a nearly linear decrease in force from 0.001 seconds to 10 seconds, which aligns with the presence of relaxation times across a wide range. This implies that different branches in the model contribute to stress relaxation at different time scales.

2. **Viscoelastic Stress:** The model simulates the artery wall using five branches, each representing a different relaxation time component. These branches likely correspond to the various biological components of the arterial wall, such as smooth muscle cells, collagen fibers, and elastin fibers .The plot shows how the total viscoelastic stress and the stress in each branch of the viscoelastic model change over time.

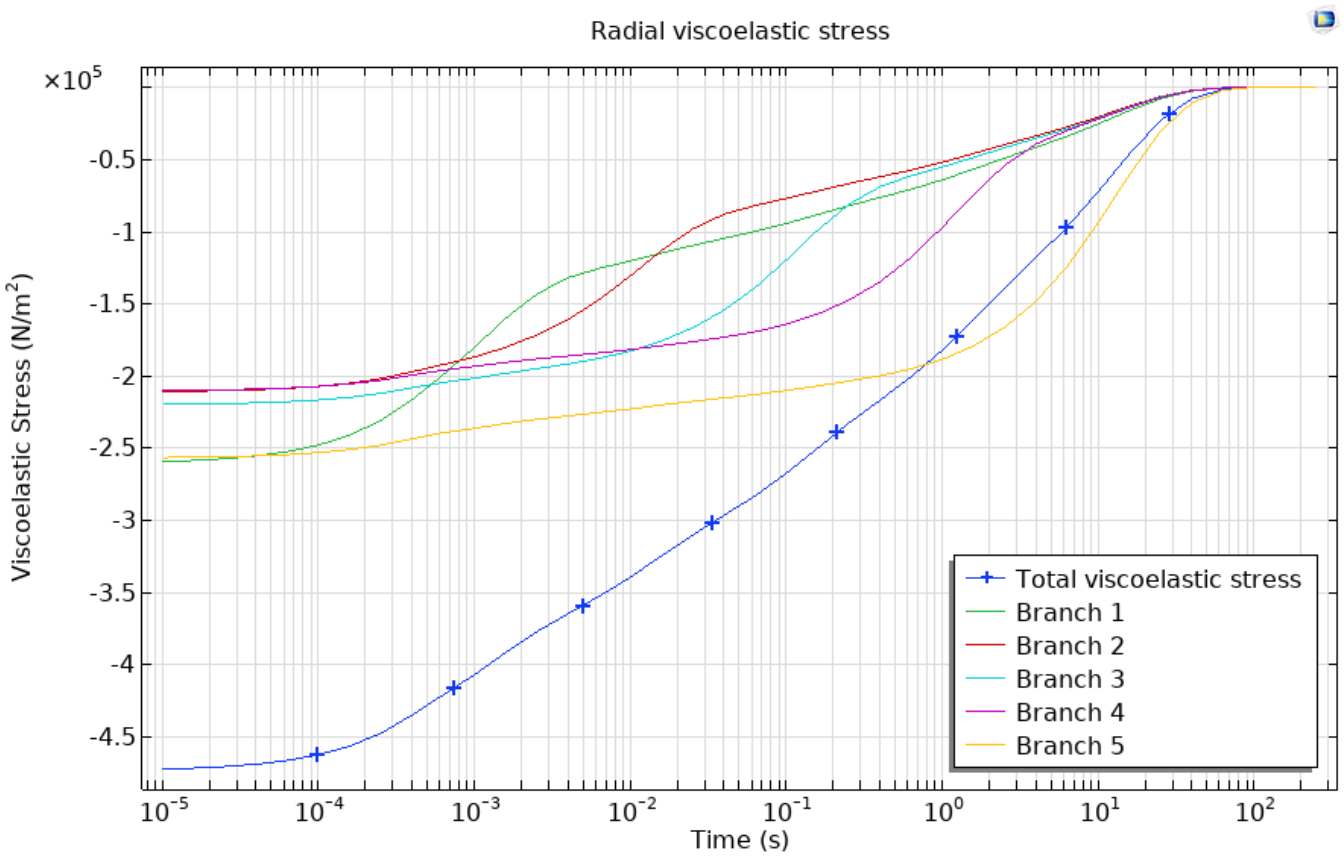


Figure 4.3.2: Viscoelastic Stress (Nm^{-2}) vs Time (sec) plot

Observation from the plot:

- When a stress or strain is applied to the material, each branch likely consists of a spring and a dashpot in parallel.
- The spring in each branch provides an immediate elastic response, similar to how a real material would initially resist deformation.

  Immediate Response: Each branch likely shows an initial jump in stress due to the spring component's elastic response, mimicking the initial resistance to deformation in the real material.
- The dashpot in each branch contributes a delayed response, with the stress gradually decreasing over time due to energy dissipation.

  Stress Relaxation: Following the initial jump, each branch shows a stress decay over time. The rate of this decay depends on the branch's relaxation time. Branches with shorter relaxation times will relax faster (steeper decay curve), while those with longer relaxation times will relax slower (shallower decay curve).

- By combining the responses from all five branches, the model simulates the overall viscoelastic behavior of the material. This includes both the immediate elastic response and the gradual stress relaxation over time.

Benefits of a Multi-Branch Model:

A single-branch model wouldn't be able to capture the complexity of real viscoelastic behavior. Compared to a single-branch model, a five-branch model offers greater flexibility in capturing the complexity of real-world viscoelastic behavior. By incorporating multiple branches with different relaxation times, the model can better represent the observed spectrum of relaxation times in actual arterial tissues. This provides a more realistic understanding of how the artery wall responds to stress and relaxes over time.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

Our project serves as a pivotal exploration into the intricate world of arterial biomechanics, offering invaluable insights and advancements in several key areas. It deepens our understanding of arterial wall behavior under varying physiological conditions. By leveraging COMSOL Multiphysics software, researchers can simulate and analyze the stress and strain distributions within arterial walls. This leads to a comprehensive understanding of how arteries deform, respond, and distribute mechanical loads, which is crucial in elucidating their mechanical properties and functionalities. The project equips researchers with advanced computational modeling skills. Through the development and validation of computational models using COMSOL Multiphysics, researchers gain proficiency in accurately representing arterial structures and simulating their mechanical behaviors. This skillset is fundamental not only for studying arterial mechanics but also for broader applications in biomechanics and biomedical engineering. It has significant clinical relevance by shedding light on the biomechanical aspects of arterial diseases. Insights gained from simulations can be translated into clinical contexts, aiding in the understanding and management of conditions such as atherosclerosis, aneurysms, and arterial stiffness. This knowledge contributes to the development of improved diagnostic techniques, treatment strategies, and patient-specific interventions in cardiovascular medicine. The project also underscores the interdisciplinary nature of biomedical research.
So, it is obvious that the project "Modeling and Simulation of Arterial Wall Mechanics by Using COMSOL Multiphysics Software" contributes significantly to our understanding of arterial biomechanics, computational modeling capabilities, clinical insights, interdisciplinary collaboration, and future research prospects in cardiovascular health and biomedical engineering.

5.2 Future Work:

The project lays a solid groundwork for future research and opens up several promising directions in the field of arterial biomechanics and computational modeling. Here are some potential future research directions:

1. Advanced Modeling Techniques: Future research could focus on enhancing the complexity and fidelity of arterial wall models. This includes incorporating more detailed anatomical features, nonlinear material properties, and fluid-structure interaction for a more realistic representation of arterial behavior.

2. Experimental Validation: validating computational models with experimental data is crucial for ensuring their accuracy and reliability. Future work could involve conducting experiments to

measure arterial mechanical properties under controlled conditions and comparing these with simulation results.

3. Patient-Specific Modeling: personalized medicine is an emerging trend in healthcare. Future research could explore the development of patient-specific arterial wall models based on individual medical imaging data. These models can help predict patient-specific responses to treatments or interventions.

4. Disease-Specific Studies: The project highlights the importance of studying arterial diseases. Future research can delve deeper into disease-specific biomechanical changes in arterial walls, such as plaque formation in atherosclerosis or aneurysm formation, to better understand disease progression and develop targeted therapies.

5. Multi-Scale Modeling: Arterial biomechanics span multiple length scales, from molecular interactions to tissue-level mechanics. Future work could integrate multi-scale modeling approaches to capture the hierarchical nature of arterial mechanics and their interplay with biological processes.

6. Dynamic Simulations: Investigating dynamic behaviors of arterial walls, such as pulsatile flow effects, arterial remodeling, and response to mechanical loading over time, can provide valuable insights into long-term cardiovascular health and disease mechanisms.

7. Optimization and Design: Computational modeling can be utilized for optimizing medical devices, such as stents or prosthetic arteries, to improve their biomechanical compatibility and long-term performance within the vascular system.

8. Clinical Translation: Translating computational models and simulations into clinical practice requires further research on integrating these tools into medical decision-making processes, treatment planning, and patient outcomes assessment.

By pursuing these future research directions, the project's impact can be extended to advance cardiovascular healthcare, improve treatment strategies, and contribute to the development of innovative biomedical technologies [18].

Studying arterial mechanics contributes to our fundamental understanding of biomechanics, tissue engineering, and materials science, fostering interdisciplinary research collaborations. Simulation-based research can lead to cost-effective solutions compared to traditional experimental methods, as it reduces the need for extensive animal studies or expensive clinical trials.

Here are some research works based on the “Arterial Wall Mechanics”

1. Blood Flow Modeling in Coronary Arteries:
Atherosclerosis is one of the main causes of cardiovascular events, namely, myocardium infarction and cerebral stroke, responsible for a great number of deaths every year worldwide. This pathology is caused by the progressive accumulation of low-density lipoproteins, cholesterol, and other

substances on the arterial wall, narrowing its lumen. To date, many hemodynamic studies have been conducted experimentally and/or numerically; however, this disease is not yet fully understood. For this reason, the research of this pathology is still ongoing, mainly, resorting to computational methods. These have been increasingly used in biomedical research of atherosclerosis because of their high-performance hardware and software. Taking into account the attempts that have been made in computational techniques to simulate realistic conditions of blood flow in both diseased and healthy arteries, the present review aims to give an overview of the most recent numerical studies focused on coronary arteries, by addressing the blood viscosity models, and applied physiological flow conditions. In general, regardless of the boundary conditions, numerical studies have been contributed to a better understanding of the development of this disease, its diagnosis, and its treatment.

Keywords: atherosclerosis; coronary arteries; hemodynamics; numerical methods [19].

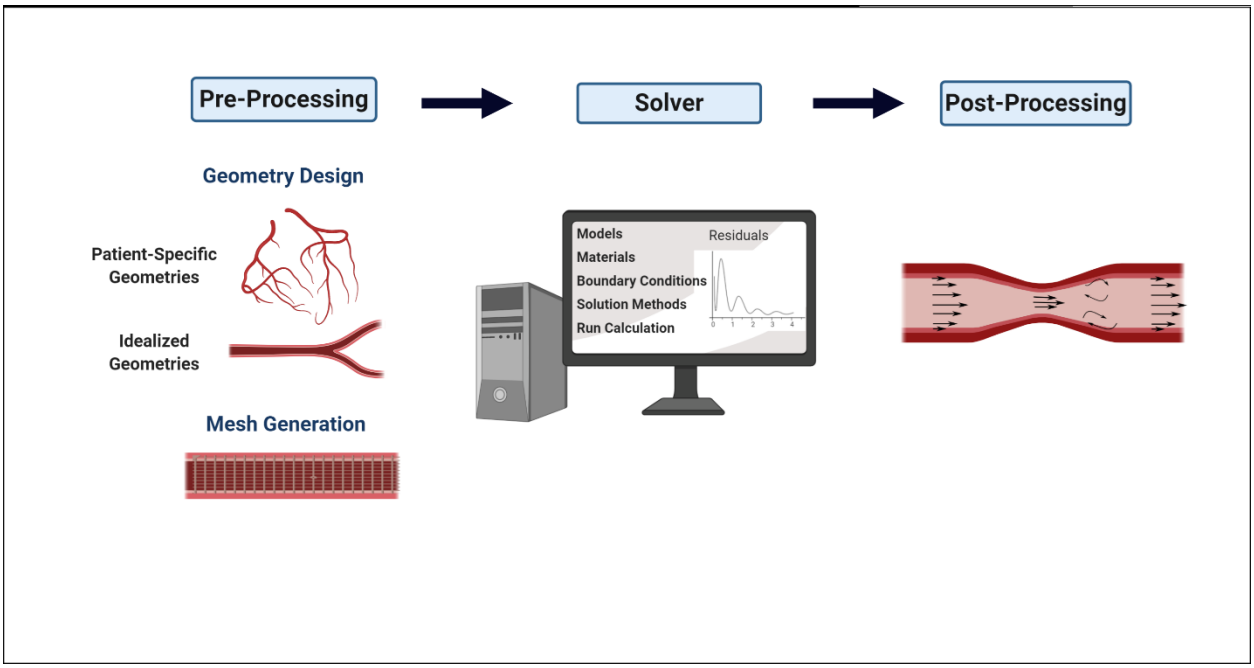


Figure 5.2.1 : Schematic diagram of blood flow modeling in coronary arteries

2. Numerical computation of blood hemodynamic through constricted human left coronary artery: Pulsatile simulations

The accumulation of plaque in the coronary artery of the human heart restricts the path of blood flow in that region and leads to Coronary Artery Disease. This study's goal is to present the pulsatile blood flow conduct through four different levels of constrictions, i.e., healthy, 25%, 50%, and 75% in human left coronary arteries.

- The left coronary artery is constructed using CT images of healthy patients. Further, we have extended this geometry to 25%, 50%, and 75% blocked arterial geometries.
- Non-Newtonian Carreau viscosity model is enforced to interpret the blood hemodynamic at different levels of blockage over the entire cardiac cycle. Also, a

comparison is made between the steady and pulsatile blood motion to interpret its effects on Coronary Artery Disease.

- It is concluded that the pulsatile flow in combination with distinct features of geometry such as branch junctions, curved sections, and flow constrictions can drive a blend of perplexing flows.
- The results will be useful for the Medical Practitioner to diagnosis, prognosis and treatment of the coronary artery diseases[20].

3. Mesh Neural Networks for SE(3)- Equivariant Hemodynamics Estimation on the Artery Wall

Computational fluid dynamics (CFD) is a valuable asset for patient-specific cardiovascular-disease diagnosis and prognosis, but its high computational demands hamper its adoption in practice. Machine-learning methods that estimate blood flow in individual patients could accelerate or replace CFD simulation to overcome these limitations. In this work, we consider the estimation of vector-valued quantities on the wall of three-dimensional geometric artery models. We employ group-equivariant graph convolution in an end-to-end SE(3)-equivariant neural network that operates directly on triangular surface meshes and makes efficient use of training data. We run experiments on a large dataset of synthetic coronary arteries and find that our method estimates directional wall shear stress (WSS) with an approximation error of 7.6% and normalised mean absolute error (NMAE) of 0.4% while up to two orders of magnitude faster than CFD. Furthermore, we show that our method is powerful enough to accurately predict transient, vector-valued WSS over the cardiac cycle while conditioned on a range of different inflow boundary conditions. These results demonstrate the potential of our proposed method as a plugin replacement for CFD in the personalized prediction of hemodynamic vector and scalar fields[21].

References:

[1] L. Bacakova et al., “The Role of Vascular Smooth Muscle Cells in the Physiology and Pathophysiology of Blood Vessels,” in Muscle Cell and Tissue - Current Status of Research Field, K. Sakuma, Ed., InTech, 2018. doi: 10.5772/intechopen.77115.

[2] D. Valdez-Jasso et al., “Viscoelastic Models for Passive Arterial Wall Dynamics,” Adv. Appl. Math. Mech., 2009.

[3] “Figure 1.2: Diagram of the major components of an elastic artery,...,” ResearchGate. Accessed: May 09, 2024. [Online]. Available: https://www.researchgate.net/figure/Diagram-of-the-major-components-of-an-elastic-artery-showing-the-three-arterial-wall_fig1_305774570 [4]Arterial tunics, Servier Medical Art, CC BY .

[5] “Structure and Function of Blood Vessels | Anatomy and Physiology II.” Accessed: May 09, 2024. [Online]. Available: https://courses.lumenlearning.com/suny-ap2/chapter/structure-and-function-of-blood-vessels/

[6] “Artery Media - an overview | ScienceDirect Topics.” Accessed: May 09, 2024. [Online]. Available: https://www.sciencedirect.com/topics/medicine-and-dentistry/artery-media

[7] “Adventitia - an overview | ScienceDirect Topics.” Accessed: May 09, 2024. [Online]. Available: https://www.sciencedirect.com/topics/immunology-and-microbiology/adventitia

[8] “Mechanical Properties of Arteries in Vivo | Circulation Research.” Accessed: May 09, 2024. [Online]. Available: https://www.ahajournals.org/doi/10.1161/01.RES.8.3.622

[9] S. A. I. G. www.applesnail.net, Anatomy overview of a human artery made for PhD in molecular genetics at the University of Maastricht in 2006. 2005. Accessed: May 09, 2024. [Online]. Available: https://commons.wikimedia.org/wiki/File:Anatomy_artery.png

[10]J.D. Humphrey, Mechanics of arterial wall: Review and directions. Critical Reviews in Biomed.Engr. 23 (1995) 1–162.

[11][12][13]H.S. Oktay, T. Kang, J. D. Humphrey and G.G. Bishop, Changes in the mechanical behavior of arteries following balloon angioplasty. In: 1991 ASME Advances in Bioengineering, New York (1991).

[14]G.A. Holzapfel, C.A.J. Schulze-Bauer and M. Stadler, Mechanics of angioplasty: Wall, balloon and stent. In: J. Casey and G. Bao (eds), Mechanics in Biology, AMD-Vol. 242/BED-Vol. 46, The American Society of Mechanical Engineers, New York (2000), pp. 141–156

[15] G. A. Holzapfel, T. C. Gasser, and M. Stadler, “A structural model for the viscoelastic behavior of arterial walls: Continuum formulation and finite element analysis,” European Journal of Mechanics - A/Solids, vol. 21, no. 3, pp. 441–463, Jan. 2002, doi: 10.1016/S0997-7538(01)01206-2.

[16] “Arterial Wall Mechanics,” COMSOL. Accessed: May 09, 2024. [Online]. Available: https://www.comsol.com/model/arterial-wall-mechanics-14499

[17] “Arterial Wall Viscoelasticity,” COMSOL. Accessed: May 09, 2024. [Online]. Available: https://www.comsol.com/model/arterial-wall-viscoelasticity-42821

[18] T.C. Gasser, C.A.J. Schulze-Bauer, E. Pernkopf, M. Stadler and G.A. Holzapfel, Rate-independent elastoplastic constitutive modeling of biological soft tissues: Part II. Percutaneous transluminal angioplasty. Submitted (2000).

[19] “Holzapfel-Gasser-Ogden Model - PolymerFEM.com.” Accessed: May 09, 2024. [Online]. Available: https://polymerfem.com/holzapfel-gasser-ogden-model/
[20] D. R. Nolan, A. L. Gower, M. Destrade, R. W. Ogden, and J. P. McGarry, “A robust anisotropic hyperelastic formulation for the modelling of soft tissue,” Journal of the Mechanical Behavior of Biomedical Materials, vol. 39, pp. 48–60, Nov. 2014, doi: 10.1016/j.jmbbm.2014.06.016.
[21] G. Holzapfel, T. Gasser, and R. Ogden, “A New Constitutive Framework for Arterial Wall Mechanics and a Comparative Study of Material Models,” J. Elasticity, vol.61, pp.1–48, 2000.